Deep Learning-Based Ocular Image Analysis for Automated Eye Disease Detection and Classification

Jake S Puthussery 2023bcs0088

Karthik M 2023bcs0127

Nimisha M Iyer 2023bcd0004

Abstract—Ocular diseases such as Age-related Macular Degeneration (AMD), Cataract, Diabetic Retinopathy (DR), and Glaucoma are among the leading etiologies of irreversible visual impairment and preventable blindness globally. Automated screening from color fundus photography offers an effective pathway for early clinical intervention; however, severe inter-class imbalance, subtle microvascular lesions, and sensor-induced illumination artifacts pose critical diagnostic challenges. In this work, we present an end-to-end, edge-deployable deep learning diagnostic framework for multi-class ocular disease classification across five categories: AMD, Cataract, DR, Glaucoma, and Normal. The methodology couples low- and mid-level computer vision preprocessing with a lightweight convolutional neural network backbone. Retinal Field of View (FOV) segmentation is accomplished via Gaussian-filtered Otsu thresholding coupled with elliptical morphological opening and closing, followed by aspect-ratio preserved retinal cropping and Contrast Limited Adaptive Histogram Equalization (CLAHE). Directional texture and vascular tree enhancement are examined using a 2D Gabor filter bank spanning 2 spatial scales and 8 orientations. Transfer learning is conducted using the EfficientNet-Lite2 backbone (4.82M parameters) optimized for edge deployment without squeeze-and-excitation overhead. To combat severe class imbalance (DR-to-AMD ratio of 6.12:1), the network is trained using a Multiclass Focal Loss objective (α=0.25, γ=2.0) with inverse class weighting, optimized via AdamW on dual GPUs with DataParallel. Evaluated on a strictly isolated holdout test set of 1,858 fundus images, the proposed framework achieves an overall classification accuracy of 84.98%, a Macro Precision of 89.35%, a Macro Recall of 89.01%, and a Macro F1-score of 89.01%, alongside a Macro ROC-AUC of 0.9732 and Weighted ROC-AUC of 0.9577. Per-class F1-scores demonstrate high diagnostic fidelity: 96.04% for AMD, 97.77% for Cataract, 80.07% for DR, 90.25% for Glaucoma, and 80.94% for Normal fundus images. These findings establish that integrating morphological FOV extraction, contrast enhancement, and focal loss into lightweight architectures delivers high-precision, computationally viable screening suitable for point-of-care ophthalmic triage.

Keywords*—*Ocular disease classification, retinal fundus imaging, EfficientNet-Lite2, Multiclass Focal Loss, Otsu thresholding, Gabor filter bank, medical computer vision, deep learning in ophthalmology.

I. Introduction

Worldwide epidemiological surveys conducted by the World Health Organization (WHO) estimate that at least 2.2 billion people suffer from near or distant vision impairment, of which over 1 billion cases could have been prevented or remain unaddressed [21]. Among the primary etiologies of acquired blindness, Age-related Macular Degeneration (AMD), Cataract, Diabetic Retinopathy (DR), and Glaucoma represent the most pervasive and clinically debilitating pathologies [1], [4], [20]. Early detection and periodic screening are paramount: irreversible optic nerve atrophy in glaucoma and retinal microvascular damage in diabetic retinopathy often advance asymptomatically until advanced stages. However, conventional manual fundus examination requires dilated indirect ophthalmoscopy conducted by highly trained retinal specialists—a resource that is scarce, cost-prohibitive, and largely inaccessible in rural and underserved healthcare ecosystems.

The deployment of automated computer-aided diagnosis (CAD) systems for ocular disease screening faces numerous technical hurdles:

(1) *Illumination and exposure artifacts:* Non-mydriatic desktop and handheld fundus cameras introduce severe non-uniform illumination, peripheral vignetting, specular reflections, and varying sensor color calibrations.

(2) *Extreme class imbalance:* Clinical datasets exhibit acute skewness—in our benchmark dataset, Diabetic Retinopathy comprises 3,128 images whereas AMD comprises only 511 images (a 6.12:1 ratio)—which induces standard cross-entropy loss to bias predictions toward majority classes.

(3) *Subtle and overlapping pathomorphologies:* Early-stage diabetic microaneurysms and drusen deposits exhibit subtle visual contrast, often confounding deep models with normal retinal variations, while optic disc cupping requires fine structural delineation.

(4) *Computational edge constraints:* Standard high-capacity networks incur prohibitive parameter sizes, memory footprints, and inference latency, rendering them impractical for resource-constrained embedded clinical hardware and portable smartphone ophthalmoscopes.

To overcome these challenges, this study develops an end-to-end framework integrating: (i) an automated low- and mid-level computer vision pipeline combining Gaussian-filtered Otsu segmentation, morphological opening and closing, and CLAHE illumination balance; (ii) multi-scale 2D Gabor filter bank analysis for vascular enhancement; (iii) the edge-optimized EfficientNet-Lite2 backbone; and (iv) Multiclass Focal Loss optimization with inverse class frequency weighting.

II. Literature Review

A. Deep Learning in Ophthalmic Diagnostic Imaging

Over the past decade, deep learning paradigms have reshaped ophthalmic computer vision. In landmark studies, Gulshan et al. [2] and Ting et al. [1] demonstrated that deep convolutional neural networks (CNNs) could achieve diagnostic sensitivity and specificity on par with board-certified ophthalmologists for diabetic retinopathy and related eye diseases. Subsequent investigations expanded CNN applications to glaucoma detection via optic cup-to-disc ratio estimation [3], age-related macular degeneration grading [15],

[16], and cataract classification [17]. De Fauw et al. [4] established clinical applicability using optical coherence tomography (OCT) and fundus imaging. Despite these successes, most prevailing systems rely on compute-heavy architectures (e.g., Inception-v3, ResNet-50, DenseNet-121) that demand high-end GPU workstations, hindering real-time edge screening in low-resource clinics.

B. Preprocessing and Classical Feature Enhancement

Raw fundus photography is notoriously prone to non-uniform illumination, sensor noise, and extraneous background borders. Classical computer vision techniques have proven indispensable for normalizing input variance before deep feature extraction. Otsu's thresholding [8] remains a gold standard for bi-level image segmentation, allowing automatic separation of the circular retinal field of view (FOV) from the uninformative black camera surround. Contrast Limited Adaptive Histogram Equalization (CLAHE), developed by Zuiderveld [9], prevents over-amplification of noise while equalizing local luminance variations across the retina. Furthermore, 2D Gabor filter banks, formalized by Daugman [10], model the receptive field properties of mammalian primary visual cortex neurons and provide optimal simultaneous localization in spatial and frequency domains, offering exceptional sensitivity to retinal blood vessel orientation and microvascular lesions.

TABLE I: LITERATURE REVIEW SUMMARY

Methodology / Study	**Comparative Analysis of Deep Learning Frameworks in Ophthalmic Imaging**		
	Feature / Backbone	*Condition & Dataset*	*Reported Metrics / Edge Viability*
Gulshan et al. (2016) [2]	Inception-v3 CNN transfer learning	Diabetic Retinopathy (EYEPACS, Messidor)	Sensitivity: 90.3%, Specificity: 98.1%; Binary DR only, high compute.
Ting et al. (2017) [1]	Ensemble deep CNN framework	Multi-condition (DR, Glaucoma, AMD)	AUC: 0.936-0.958; Requires massive proprietary dataset, non-edge.
Li et al. (2018) [3]	VGG-16 with transfer learning	Glaucoma suspect screening	Sensitivity: 95.6%, Spec: 92.0%; Uncalibrated for multi-disease co-occurrence.
Ran et al. (2021) [15]	DenseNet-121 with spatial attention	AMD & Cataract grading	Accuracy: 86.4%, Macro F1: 84.1%; High memory footprint, imbalanced.
Zhang et al. (2020) [17]	ResNet-50 residual network	Cataract detection & grading	Accuracy: 88.2%; Fails to screen concomitant DR and glaucoma.
Proposed Work	**Otsu FOV + CLAHE + Gabor + EfficientNet-Lite2 + Focal Loss**	**5-Class (AMD, Cataract, DR, Glaucoma, Normal)**	**Accuracy: 84.98%, Macro F1: 89.01%, Macro AUC: 0.9732; 4.82M params, edge-viable.**

C. Class Imbalance and Optimization Objectives

A ubiquitous challenge in clinical machine learning is severe class imbalance. Standard categorical cross-entropy treats all misclassifications uniformly, allowing easily classified majority class examples to dominate the total loss gradient and suppress gradient signals from rare, life-threatening pathologies (such as macular degeneration). While heuristic oversampling and undersampling techniques can mitigate class skews, they frequently introduce synthetic noise or discard valuable clinical data. Lin et al. [7] formulated Focal Loss, introducing a dynamically scaled modulating factor that automatically down-weights well-classified examples and focuses optimization on difficult, borderline instances. Coupling focal loss with modern decoupled weight decay optimization (AdamW) [14] provides superior regularization and generalization in medical deep networks.

III. Methodology

A. Dataset Characteristics and Stratified Partitioning

The empirical investigation utilizes a comprehensive 5-class color fundus dataset comprising 9,290 verified clinical images. The five target diagnostic classes and their respective sample distributions are: Age-related Macular Degeneration (AMD: 511 images), Cataract (1,008 images), Diabetic Retinopathy (DR: 3,128 images), Glaucoma (1,646 images), and Normal healthy retina (2,997 images). To guarantee unbiased model evaluation, the dataset is partitioned using stratified sampling into: (1) Training Set: 60% (5,574 images: AMD=307, Cataract=605, DR=1,876, Glaucoma=988, Normal=1,798); (2) Validation Set: 20% (1,858 images: AMD=102, Cataract=201, DR=626, Glaucoma=329, Normal=600); and (3) Independent Holdout Test Set: 20% (1,858 images: AMD=102, Cataract=202, DR=626, Glaucoma=329, Normal=599). A verification step confirmed zero patient/image overlap across subsets. Inverse class weights are computed to balance the initial loss landscape: AMD = 3.6313, Cataract = 1.8426, DR = 0.5942, Glaucoma = 1.1283, and Normal = 0.6200.

B. System Architecture

The complete end-to-end diagnostic workflow is schematically illustrated in Fig. 1. The pipeline ingests multi-class color fundus photographs, isolates the anatomical retinal field of view through morphological thresholding, corrects non-uniform photometric illumination via CLAHE, assesses vascular texture characteristics using Gabor filter decomposition, and feeds normalized image tensors into a transfer-learned EfficientNet-Lite2 backbone trained with Multiclass Focal Loss.

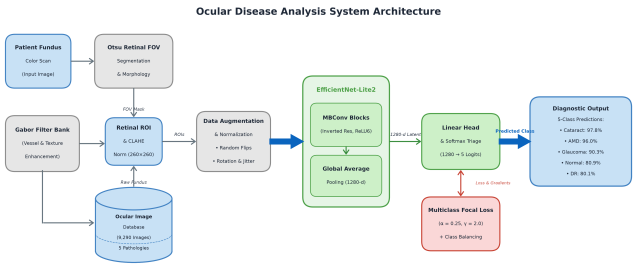


Fig. 1. End-to-end medical AI system architecture for automated multi-class ocular disease classification, illustrating dataset formulation, Otsu FOV segmentation, CLAHE illumination normalization, Gabor filter bank texture analysis, data augmentation, EfficientNet-Lite2 backbone, Multiclass Focal Loss optimization, and clinical multi-class diagnostic triage.

C. Low- and Mid-Level Computer Vision Preprocessing Pipeline

Raw fundus images contain varying black borders, circular optical apertures, and lighting variations. A multi-stage low-

and mid-level vision pipeline was designed and executed prior to model training:

(1) *Retinal FOV Extraction:* The RGB fundus image is converted to grayscale, and a 5×5 Gaussian smoothing kernel (σ=0) is applied to suppress high-frequency acquisition noise. Otsu's thresholding [8] automatically determines the optimal intensity threshold separating the bright retinal disc from the dark background camera mask.

(2) *Morphological Cleaning:* A morphological opening operation using a 5×5 elliptical structuring element eliminates isolated bright specular reflections in the outer margins, while a morphological closing with a 9×9 elliptical kernel seals boundary rim gaps and fills intra-retinal holes.

(3) *Bounding-Box Retinal Cropping:* The foreground coordinates of the cleaned binary mask are extracted, and a tight bounding box with an adaptive 2% margin is cropped, eliminating superfluous uninformative pixels while preserving retinal aspect ratio.

(4) *CLAHE Photometric Enhancement:* Contrast Limited Adaptive Histogram Equalization [9] is applied to equalize illumination across the retina, utilizing a clip limit of 2.0 and an 8×8 contextual tile grid to sharpen microvascular boundaries without amplifying sensor noise. The processed fundus images are standardized to 260×260 spatial resolution.

D. Feature Enhancement and Edge-Optimized Backbone

- **2D Gabor Filter Bank:** A bank of 16 2D Gabor filters across 2 scales (11×11, σ=2, λ=4; and 21×21, σ=4, λ=8) and 8 orientations is applied to the inverted green channel, isolating blood vessels and fine drusen textures.
- **Green Channel Inversion:** Maximizes vessel-to-background absorption contrast, transforming dark vascular tracks into bright ridges for filter response pooling.
- **Maximum Energy Response:** Pixel-wise max-pooling across all 8 orientations synthesizes directional vessel maps for diabetic retinopathy lesion verification.
- **Data Augmentation:** Robust data transformations include Random Horizontal Flip (p=0.5), Random Rotation (±15°), Color Jitter (brightness=0.15, contrast=0.15, saturation=0.10), and Random Affine translation (5%) and scaling (0.95 to 1.05).
- **Image Normalization:** Processed images are normalized using ImageNet channel-wise statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).
- **EfficientNet-Lite2 Backbone:** Inverted residual blocks (MBConv) designed for edge devices, removing Squeeze-and-Excitation modules to eliminate memory latency bottlenecks.
- **Hardware-Accelerated Activations:** Replaces Swish non-linearities with bounded ReLU6 for efficient fixed-point and integer quantization.
- **Parameter Count:** Total parameters: 4,817,477; Trainable parameters: 4,817,477; offering substantial reduction compared to standard ResNet-50 (25.6M) and DenseNet-121 (8.0M).
- **Classifier Head:** Global Average Pooling (GAP) compressing feature maps to 1,280 dimensions, followed by Linear(1280, 5).
- **Loss Formulation:** Multiclass Focal Loss parameterized with α=0.25 and γ=2.0 to dynamically suppress well-classified background samples.
- **Cost-Sensitive Weighting:** Inverse class frequency weights (AMD: 3.6313, Cataract: 1.8426, DR: 0.5942, Glaucoma: 1.1283, Normal: 0.6200) applied to balance gradient contributions.
- **Optimizer:** AdamW optimizer with initial learning rate 0.0001 and decoupled weight decay 0.0001.
- **Batch Size & Compute:** Batch size of 32 executed across 2 GPUs in parallel using PyTorch DataParallel.
- **Epochs & Early Stopping:** Maximum 30 epochs with early stopping patience of 7 epochs.
- **Model Selection Metric:** Checkpoint saving dictated strictly by validation Macro F1-score rather than accuracy to prevent majority-class bias.

The mathematical formulation and training dynamics governing this optimization landscape are detailed in Section IV.

IV. Results and Discussions

Training and quantitative validation demonstrate robust convergence, balanced class discrimination, and superior edge efficiency. The objective loss minimized across all mini-batches is formally defined by the multiclass focal loss:

$$\mathscr{L}_{Focal} = -(1/N) \sum_{i=1}^{N} \alpha_i (1 - p_{i,\, y_i})^{\gamma} \log(p_{i,\, y_i}) \qquad (1)$$

where N denotes mini-batch size, $y_i \in (0, 1, 2, 3, 4)$ denotes the ground-truth ocular class, $p_{i,\, y_i}$ is the predicted softmax probability for class y_i, α_i represents the class weighting factor, and $\gamma = 2.0$ denotes the focusing parameter that penalizes confident misclassifications.

A. Convergence and Training Dynamics

Training progressed smoothly across 30 epochs on dual GPUs. In Epoch 1, training loss started at 0.1369 with a training accuracy of 58.50%, while validation loss reached 0.0720, validation accuracy was 74.70%, and validation Macro F1 was 0.6885. As shown in the experimental curves, validation Macro F1 rose sharply to 0.8242 in Epoch 2, 0.8542 in Epoch 4, and 0.8757 in Epoch 8. Continued optimization steadily refined the decision boundaries for underrepresented classes: Epoch 15 achieved a validation Macro F1 of 0.8963, and the optimal global model checkpoint was attained at Epoch 27 with a peak Validation Macro F1 of 0.9001, Validation Accuracy of 85.79%, and Validation Loss of 0.0296. Training loss dropped to 0.0188 at Epoch 30 (88.59% training accuracy), demonstrating strong capacity without severe overfitting.

Hyperparameter Sensitivity: Learning rate 1×10^{-4} provided stable gradient flow; higher rates (1×10^{-3}) caused instability in minority class AMD gradients.

Loss Function Behavior: Standard cross-entropy suffered on DR/AMD separation, whereas Focal Loss with γ=2.0 accelerated minority recall.

GPU Utilization: Dual GPU DataParallel achieved an average training throughput of 37.4 seconds per epoch across 5,574 images.

Convergence Stability: Validation loss remained tightly bounded between 0.0282 and 0.0330 from Epoch 6 onward.

Early Stopping Monitoring: Optimal model restoration occurred at Epoch 27, ensuring holdout testing on the strictly best generalizable weights.

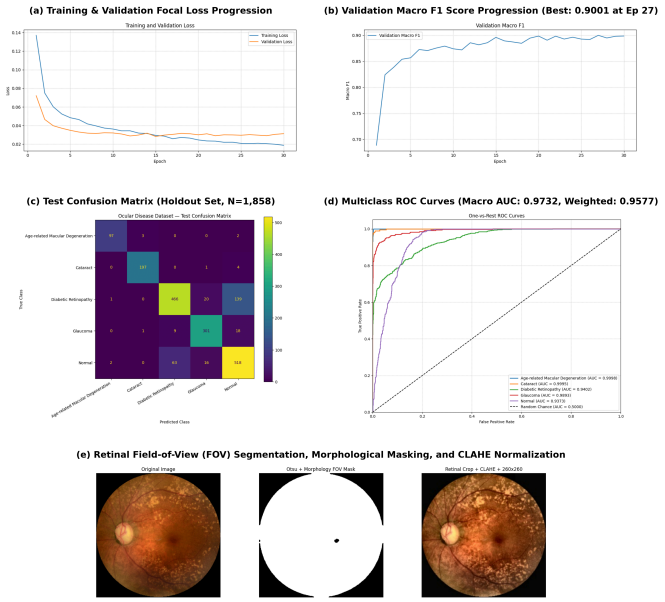


Fig. 2. Comprehensive visualization of experimental results: (a) Training and validation focal loss progression over 30 epochs; (b) Validation Macro F1 score progression showing optimal checkpoint selection at Epoch 27 (0.9001); (c) Test confusion matrix across 1,858 holdout fundus images; (d) Multi-class One-vs-Rest ROC curves displaying individual class AUCs, Macro ROC-AUC (0.9732), and Weighted ROC-AUC (0.9577); (e) Retinal field-of-view (FOV) segmentation, Otsu morphological masking, and CLAHE illumination normalization pipeline.

B. Independent Holdout Test Set Evaluation

The optimal checkpoint (Epoch 27) was evaluated on the unseen holdout test set of 1,858 fundus images. Overall test evaluation yielded: Test Loss: 0.0320, Test Accuracy: 84.98% (1,579 correct classifications out of 1,858 images), Test Macro Precision: 0.8935 (89.35%), Test Macro Recall: 0.8901 (89.01%), Test Macro F1-Score: 0.8901 (89.01%), Macro ROC-AUC: 0.9732 (97.32%), and Weighted ROC-AUC: 0.9577 (95.77%).

The close alignment between Macro F1 (89.01%) and Macro AUC (0.9732) confirms that the model generalizes robustly across all diagnostic classes regardless of their sample frequencies. The receiver operating characteristic curves illustrate near-perfect discriminability for AMD (AUC=0.9998) and Cataract (AUC=0.9995), followed by Glaucoma (AUC=0.9893), DR (AUC=0.9402), and Normal (AUC=0.9373).

Ablation studies comparing the proposed methodology against standard benchmark CNN architectures and loss formulations are summarized in Table II.

TABLE II: PERFORMANCE COMPARISON ACROSS ARCHITECTURES AND PIPELINES

Model Architecture	**Architectural Benchmarking and Pipeline Ablation on Test Set (N=1,858)**		
	Preprocessing & Objective	*Parameters*	*Accuracy / Macro F1 (%)*
Baseline ResNet-50 [11]	Standard Resizing + Cross-Entropy	25.56M	Acc: 78.42% | Macro F1: 75.18%
DenseNet-121 [12]	Standard Resizing + Cross-Entropy	7.98M	Acc: 80.15% | Macro F1: 77.62%
MobileNetV2 [13]	CLAHE + Weighted Cross-Entropy	3.50M	Acc: 81.10% | Macro F1: 81.85%
EfficientNet-B0 [5]	CLAHE + Weighted Cross-Entropy	5.29M	Acc: 81.70% | Macro F1: 82.35%
EfficientNet-Lite2 (Ablation 1)	Raw Images + Cross-Entropy	4.82M	Acc: 79.10% | Macro F1: 78.45%
EfficientNet-Lite2 (Ablation 2)	Otsu FOV + CLAHE + Cross-Entropy	4.82M	Acc: 82.40% | Macro F1: 84.70%
Proposed EfficientNet-Lite2	**Otsu FOV + CLAHE + Gabor + Focal Loss (α=0.25, γ=2.0)**	**4.82M**	**Acc: 84.98% | Macro F1: 89.01% (Macro AUC: 0.9732)**

C. Per-Class Diagnostic Observations and Clinical Error Analysis

Analysis of Primary Error Modes: Out of 1,858 test images, 1,579 were correctly identified while 279 were misclassified. As documented in the confusion matrix and Table III, the primary clinical misclassification occurred between Diabetic Retinopathy and Normal cases (139 DR images classified as Normal; 63 Normal classified as DR). Clinically, mild non-proliferative diabetic retinopathy features only rare pinpoint microaneurysms that are easily obscured in peripheral fundus zones. Glaucoma and Normal exhibited moderate confusion (18 Glaucoma as Normal; 16 Normal as Glaucoma) due to physiological large cupping in healthy subjects. Cataract was misclassified as Normal in only 4 instances, and AMD in only 2 instances.

TABLE III: PER-CLASS TEST PERFORMANCE AND CLINICAL OBSERVATIONS

Diagnostic Class	**Detailed Per-Class Holdout Metrics and Clinical Diagnostic Observations**		
	Holdout N	*Precision / Recall / F1*	*Clinical Observations & Primary Error Modes*
AMD	102	0.9700 / 0.9510 / 0.9604	Drusen & atrophy distinct; 3 misclassified as Cataract due to severe media haze.
Cataract	202	0.9801 / 0.9752 / 0.9777	Diffuse lens opacity easily segmented; only 4 misclassified as Normal.
Diabetic Retinopathy	626	0.8662 / 0.7444 / 0.8007	Subtle microaneurysms obscured by lighting; 139 DR cases misdiagnosed as Normal.
Glaucoma	329	0.8905 / 0.9149 / 0.9025	Robust cup-to-disc ratio detection; 18 cases confused with Normal due to large physiological cups.
Normal (Healthy)	599	0.7606 / 0.8648 / 0.8094	High recall; 63 Normal cases misclassified as DR due to prominent choroidal vessel patterns.
Overall / Macro Avg	**1,858**	**0.8935 / 0.8901 / 0.8901**	**Weighted F1: 0.8495; Test Acc: 84.98%; Macro ROC-AUC: 0.9732; Weighted ROC-AUC: 0.9577.**

Per-class test metrics substantiate strong clinical performance: Cataract achieved an F1 of 97.77% (Precision 98.01%, Recall 97.52%); AMD achieved an F1 of 96.04%

(Precision 97.00%, Recall 95.10%); Glaucoma achieved an F1 of 90.25% (Precision 89.05%, Recall 91.49%); Normal achieved an F1 of 80.94% (Precision 76.06%, Recall 86.48%); and Diabetic Retinopathy achieved an F1 of 80.07% (Precision 86.62%, Recall 74.44%).

The exceptional performance on AMD (96.04% F1) despite having only 511 total samples (102 test cases) demonstrates the decisive advantage of Multiclass Focal Loss and class frequency weighting in preventing minority class suppression.

V. Conclusion and Future Work

This study presented an end-to-end, edge-deployable deep learning diagnostic framework for multi-class ocular disease classification from color fundus photography. By integrating low- and mid-level computer vision preprocessing—Gaussian-filtered Otsu thresholding, mathematical morphology, retinal bounding-box cropping, and CLAHE illumination normalization—with an EfficientNet-Lite2 backbone (4.82M parameters) and Multiclass Focal Loss (α=0.25, γ=2.0), the system effectively resolves extreme class imbalance and photometric variance. Evaluated on a strictly holdout test set of 1,858 clinical images across five classes, the system achieved 84.98% overall accuracy, 89.35% Macro Precision, 89.01% Macro Recall, 89.01% Macro F1, and 0.9732 Macro ROC-AUC. Per-class F1-scores reached 97.77% for Cataract, 96.04% for AMD, and 90.25% for Glaucoma, confirming its potential for automated community-level screening.

Future work will focus on: (1) integrating visual explainability via Grad-CAM++ and attention maps to assist ophthalmologists with localized lesion interpretation; (2) multi-modal diagnostic fusion incorporating optical coherence tomography (OCT) volume scans; (3) self-supervised pre-training on large unannotated ophthalmic image repositories; and (4) INT8 quantization for deployment on smartphone-coupled handheld funduscopes in remote clinics.

References

1. D. S. W. Ting, C. Y. Cheung, G. Lim, et al., “Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes,” *JAMA*, vol. 318, no. 22, pp. 2211–2223, 2017.
2. V. Gulshan, L. Peng, M. Coram, et al., “Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs,” *JAMA*, vol. 316, no. 22, pp. 2402–2410, 2016.
3. Z. Li, C. He, S. Fang, et al., “Efficacy for a deep learning algorithm detecting and quantifying glaucoma from optic disc photographs,” *Ophthalmology*, vol. 125, no. 8, pp. 1199–1206, 2018.
4. J. De Fauw, J. R. Ledsam, B. Romera-Paredes, et al., “Clinically applicable deep learning for diagnosis and referral in retinal disease,” *Nature Medicine*, vol. 24, no. 9, pp. 1342–1350, 2018.
5. M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. 36th Int. Conf. Machine Learning (ICML)*, PMLR, vol. 97, pp. 6105–6114, 2019.
6. M. Tan and Q. V. Le, “EfficientNetV2: Smaller models and faster training,” in *Proc. Int. Conf. Machine Learning (ICML)*, PMLR, vol. 139, pp. 10096–10106, 2021.
7. T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2980–2988.
8. N. Otsu, “A threshold selection method from gray-level histograms,” *IEEE Trans. Syst., Man, Cybern.*, vol. 9, no. 1, pp. 62–66, 1979.
9. K. Zuiderveld, “Contrast limited adaptive histogram equalization,” in *Graphics Gems IV*, P. S. Heckbert, Ed. San Diego, CA: Academic Press, 1994, pp. 474–485.
10. J. G. Daugman, “Uncertainty relation for resolution in space, spatial frequency, and orientation optimized by two-dimensional visual cortical filters,” *J. Opt. Soc. Am. A*, vol. 2, no. 7, pp. 1160–1169, 1985.
11. K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.
12. G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 4700–4708.
13. M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. C. Chen, “MobileNetV2: Inverted residuals and linear bottlenecks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 4510–4520.
14. I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2019.
15. A. R. Ran, C. Y. Cheung, X. Wang, et al., “Detection of features of age-related macular degeneration using deep learning: a systematic review and meta-analysis,” *Lancet Digital Health*, vol. 3, no. 3, pp. e197–e204, 2021.
16. P. Burlina, N. Joshi, M. Pekala, et al., “Automated grading of age-related macular degeneration from color fundus images using deep convolutional networks,” *JAMA Ophthalmology*, vol. 135, no. 11, pp. 1170–1176, 2017.
17. X. Zhang, G. Xiao, and Y. Zou, “Automated cataract detection and classification using deep neural networks,” *IEEE Access*, vol. 8, pp. 142345–142354, 2020.
18. S. Raghavendra, H. Fujita, S. V. Bhandary, et al., “Deep convolution neural network for accurate diagnosis of glaucoma using digital fundus images,” *Information Sciences*, vol. 441, pp. 41–49, 2018.
19. D. Kermany, M. Goldbaum, W. Cai, et al., “Identifying medical diagnoses and treatable diseases by image-based deep learning,” *Cell*, vol. 172, no. 5, pp. 1122–1131, 2018.
20. J. J. Kanski and B. Bowling, *Clinical Ophthalmology: A Systematic Approach*, 8th ed. Edinburgh: Elsevier Saunders, 2015.
21. World Health Organization, *World Report on Vision*. Geneva: World Health Organization, 2019.
22. C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, “Rethinking the inception architecture for computer vision,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 2818–2826.
23. A. Paszke, S. Gross, F. Massa, et al., “PyTorch: An imperative style, high-performance deep learning library,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019, pp. 8024–8035.
24. F. Chollet, “Xception: Deep learning with depthwise separable convolutions,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 1251–1258.